

Continuous Assessment II

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Question 1

0.0.1 (a)

We made various assumptions regarding distributions and distances. For example we assumed that the distance from the source to the observer was much much larger than the volume of the source. If we were to approximate the source as a n dimensional sphere with radius a , $\vec{x}'$ being a vector to a point on that n-sphere and the observer is a distance $\vec{x}$, with $|\vec{x}| = r$ from the centre of the source then we have $r \gg a \geq \vec{x}$.

We also make the assumption that $\frac{a}{c} |\frac{1}{\rho} \frac{\partial \rho}{\partial t}| \ll 1$. Which has the implication on the maxwell frequency ($\rho_\omega(x) = 0, \forall \omega > \omega_{max}$) that $\frac{a}{c} |\omega_{max}| \ll 1 \leftrightarrow \lambda_{min} \gg a$. This assumption significantly simplifies our expressions. It implies that

$$\frac{\vec{x}' \cdot \vec{n}}{c} < \frac{a}{c} \quad (1)$$

So we can make an approximation and taylor expand about $\frac{\vec{x}' \cdot \vec{n}}{c} = 0$ in expressions off ρ and $\vec{j}$.

We also made a final assumption that the wavelength is a lot smaller than the distant once to the observer which is given by

$$\lambda \ll r \quad (2)$$

From lectures we have

$$\vec{A}(t, \vec{x}) = \frac{1}{c} \int d^3 \vec{x} \frac{1}{|\vec{x} - \vec{x}'|} \vec{j} \left(t - \frac{|\vec{x} - \vec{x}'|}{c}, \vec{x} \right) \quad (3)$$

$$\vec{A}(t, \vec{x}) = \frac{1}{c} \frac{1}{|\vec{x}|} \int d^3 \vec{x} \frac{1}{|1 - \frac{\vec{x}' \cdot \vec{x}}{r}|} \vec{j} \left(t - \frac{|\vec{x} - \vec{x}'|}{c}, \vec{x} \right) \quad (4)$$

We can taylor expand for small $\vec{x}'$ using $(1+x)^\alpha \approx 1 + \alpha x$ and $\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (x-0)^k$ to get

$$\vec{A}(t, \vec{x}) = \frac{1}{rc} \int d^3 \vec{x} \left(1 + \frac{\vec{x}' \cdot \vec{x}}{r^2} + \dots \right) \left[\vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) + \left(\frac{\vec{x}' \cdot \vec{x}}{r} \right) \dot{\vec{j}} \left(t - \frac{r}{c}, \vec{x} \right) + \dots \right] \quad (5)$$

We can apply the Electric Dipole approximation by dropping $\mathcal{O}\left(\frac{a}{\lambda_{min}}\right)$ (keepin only leading order).

$$\Rightarrow \vec{A}^{ED} = \frac{1}{rc} \int d^3 \vec{x}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) \quad (6)$$

$$\vec{B}^{ED} = \vec{\nabla} \times \vec{A}^{ED} = \frac{1}{rc} \left[\vec{\nabla} \times \int d^3 \vec{x}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) \right] \quad (7)$$

$$(8)$$

We can now use the continuity equation

$$\dot{\rho} + \vec{\nabla} \cdot \vec{j} = 0 \quad (9)$$

$$\partial_k (j_k x_l) = (\partial_k d_k) x_l + j_k \partial_k x_l = (\partial_k j_k) x_l + j_l \quad (10)$$

$$\vec{j} = \partial_k (j_k \vec{x}) - (\vec{\nabla} \cdot \vec{j}) \vec{x} \quad (11)$$

With $\vec{x}'$

$$\vec{j}(t, \vec{x}') = \partial'_k (j_k(t, \vec{x}) \vec{x}') - \vec{\nabla}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) \vec{x}' \quad (12)$$

$$\implies \vec{A}_{ret}(t, \vec{x}) \approx \frac{1}{rc} \int d^3 \vec{x}' \left[\partial'_k (j_k(t, \vec{x}) \vec{x}') - \vec{\nabla}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) \vec{x}' \right] + \dots \quad (13)$$

$$\text{we note } -\vec{\nabla}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) = \dot{\rho} \left(t - \frac{r}{c}, \vec{x}' \right) \text{ and the other term vanishes as its a total derivative.} \quad (14)$$

$$\vec{A}_{ret} = \frac{1}{rc} \int d^3 \vec{x}' \dot{\rho} \left(t - \frac{r}{c}, \vec{x}' \right) + \dots \quad (15)$$

$$= \frac{1}{rc} \frac{d}{dt} \int d^3 \vec{x}' \rho \left(t - \frac{r}{c}, \vec{x}' \right) = \frac{1}{rc} \dot{d} \left(t - \frac{r}{c} \right) \quad (16)$$

$$\implies B^{ED} = \vec{\nabla} \times \vec{A}^{ED} = \left[\vec{\nabla} \times \frac{1}{rc} \int d^3 \vec{x}' \vec{j} \left(t - \frac{r}{c}, \vec{x}' \right) \right] = \vec{\nabla} \times \vec{A}^{ED} = \left[\vec{\nabla} \times \frac{1}{rc} \dot{d} \left(t - \frac{r}{c} \right) \right] \quad (17)$$

$$(18)$$

Now we need to do some vector algebra We have something in the form where we can use the following vector identity

$$\vec{\nabla} \times AB = \vec{\nabla} A \times B + A (\vec{\nabla} \times B) \quad (19)$$

$$\implies B^{ED} = \left(\vec{\nabla} \frac{1}{rc} \right) \times \dot{d} + \frac{1}{rc} (\vec{\nabla} \times \dot{d}) \quad (20)$$

$$= -\frac{\hat{n}}{r^2 c} \times \dot{d} + \frac{1}{rc} (\epsilon_{ijk} \partial_j \dot{d}_k) \quad (21)$$

$$= -\frac{\hat{n}}{r^2 c} \times \dot{d} + \frac{1}{rc} (\epsilon_{ijk} \ddot{d} \cdot \partial_j \left(t - \frac{r}{c} \right)) \quad (22)$$

$$= -\frac{\hat{n}}{r^2 c} \times \dot{d} + \frac{1}{rc} \left(\epsilon_{ijk} \ddot{d} \cdot -\frac{1}{c} \partial_j r \right) \quad (23)$$

$$= -\frac{\hat{n}}{r^2 c} \times \dot{d} + \frac{1}{rc} \left(\epsilon_{ijk} \ddot{d} \cdot -\frac{1}{c} \hat{n} \right) \quad (24)$$

$$(25)$$

$$B^{ED} = \frac{1}{r^2 c} \vec{n} \times \dot{d} \left(t - \frac{r}{c} \right) - \frac{1}{rc^2} \vec{n} \times \ddot{d} \left(t - \frac{r}{c} \right) \quad (26)$$

$$\text{in far field } B^{ED} \approx -\frac{1}{rc^2} \vec{n} \times \ddot{d} \left(t - \frac{r}{c} \right) \quad (27)$$

(b)

We have the following expression for the total power

$$\mathcal{P} = \int_{S^2} d^2 \vec{r} \cdot S \quad (28)$$

So we need to find the Poynting vector

$$S = \frac{c}{4\pi} E \times B = \frac{c}{4\pi} |E| |B| \sin(\theta) \hat{n} \quad (29)$$

$$\text{We know } B \perp E \implies \sin \theta = 1 \text{ and } |B| = |E| \quad (30)$$

$$\implies S = \frac{c}{4\pi} |B|^2 \hat{n} \approx \frac{c}{4\pi} |B^{ED} + B^{MD}|^2 \hat{n} = \frac{c}{4\pi} \left(|B^{ED}|^2 + |B^{MD}|^2 + 2|B^{ED}| \cdot |B^{MD}| \right) \hat{n} \quad (31)$$

$$(32)$$

We can Split squared terms into one copy in terms of the electric field scaled by $\sin \frac{\pi}{2}$ and we can use the expression for B^{ED} obtained earlier. From lecture notes we have

$$\vec{B}^{MD} = \vec{\nabla} \times A^{MD} = \frac{1}{r^2 c^2} \hat{n} \times \left[\hat{n} \times \dot{\vec{m}} \left(t - \frac{r}{c} \right) \right] + \frac{1}{rc^3} \hat{n} \times \left[\hat{n} \times \ddot{\vec{m}} \left(t - \frac{r}{c} \right) \right] \quad (33)$$

$$\text{where we see the second term dominates in the far field} \quad (34)$$

$$\implies \vec{B}^{MD} = \frac{1}{rc^3} \hat{n} \times \left[\hat{n} \times \ddot{\vec{m}} \left(t - \frac{r}{c} \right) \right] \quad (35)$$

$$(36)$$

Using these we get

$$S = \frac{c}{4\pi} \left[|B^{ED}| |E^{ED}| \sin \left(\frac{\pi}{2} \right) + |B^{MD}| |E^{MD}| \sin \left(\frac{\pi}{2} \right) + 2 \left(-\frac{1}{rc^2} \vec{n} \times \ddot{\vec{d}} \left(t - \frac{r}{c} \right) \right) \frac{1}{rc^3} \hat{n} \times \left[\hat{n} \times \ddot{\vec{m}} \left(t - \frac{r}{c} \right) \right] \right] \hat{n} \quad (37)$$

$$= \frac{c}{4\pi} \left[E^{ED} \times B^{ED} + E^{MD} \times B^{MD} \right] - \frac{1}{2\pi r^2 c^4} \ddot{\vec{d}} \left(\hat{n} \times \ddot{\vec{m}} \right) \hat{n} \quad (38)$$

$$= S^{ED} + S^{MD} + \frac{1}{2\pi r^2 c^4} \hat{n} \cdot \left(\ddot{\vec{d}} \times \ddot{\vec{m}} \right) \hat{n} \quad (39)$$

$$S = S^{ED} + S^{MD} + \frac{1}{2\pi r^2 c^4} |\ddot{\vec{d}} \times \ddot{\vec{m}}| \cos \theta \hat{n} \quad (40)$$

Now we can find the power

$$\mathcal{P} = \cdot \left[S^{ED} + S^{MD} + \frac{1}{2\pi r^2 c^4} |\ddot{\vec{d}} \times \ddot{\vec{m}}| \cos \theta \hat{n} \right] \quad (41)$$

$$\mathcal{P} = \int_{S^2} d^2 \vec{r} S^{ED} + \int_{S^2} d^2 \vec{r} S^{MD} + \int_{S^2} d^2 \vec{r} \frac{1}{2\pi r^2 c^4} |\ddot{\vec{d}} \times \ddot{\vec{m}}| \cos \theta \hat{n} \quad (42)$$

$$\mathcal{P} = \mathcal{P}^{ED} + \mathcal{P}^{MD} + \frac{|\ddot{\vec{d}} \times \ddot{\vec{m}}|}{2\pi c^4} \int_0^{2\pi} d\phi \int_0^\pi \int_0^R dr \frac{r^2}{r^2} \sin \theta \hat{n} \cdot \cos \theta \hat{n} d\theta \quad (43)$$

$$= \mathcal{P}^{ED} + \mathcal{P}^{MD} + \frac{|\ddot{\vec{d}} \times \ddot{\vec{m}}|}{c^4} \int_0^\pi R \sin \theta d(\sin \theta) \quad (44)$$

$$\implies \mathcal{P} = \mathcal{P}^{ED} + \mathcal{P}^{MD} \quad (45)$$

As required

Question 2

Given

$$P(t) = \frac{2}{3} \frac{q^2}{c} \left[\gamma^6 \left(\dot{\vec{\beta}}^2 - \left(\vec{\beta} \times \dot{\vec{\beta}} \right)^2 \right) \right]_{ret} \quad (46)$$

$$\text{given } \vec{\beta}, \dot{\vec{\beta}} \text{ differ by } \theta \implies P(t) = \frac{2q^2}{3c} \left[\gamma^6 \left(\dot{\vec{\beta}}^2 - \vec{\beta} \dot{\vec{\beta}}^2 \sin^2 \theta \right) \right]_{ret} \quad (47)$$

$$\text{We can assume we aren't moving at relativistic speeds i.e } \vec{\beta} \ll 1 \text{ so } \implies \gamma \approx 1 \quad (48)$$

$$P(t) \approx \frac{2q^2}{3c} \left[\dot{\vec{\beta}}^2 \left(1 - \vec{\beta}^2 \sin^2 \theta \right) \right]_{ret} \quad (49)$$

$$\vec{\beta}^2 \sin^2 \theta \ll 1 \quad (50)$$

$$P(t) \approx \frac{2q^2}{3c} \dot{\vec{\beta}}_{ret}^2 = \frac{2q^2}{3c} \left(\frac{\ddot{\vec{x}}}{c^2} \right)_{ret} = \frac{2}{3c^3} \left(q \ddot{\vec{x}} \right)_{ret} \quad (51)$$

$$\text{Now we can compare with the elementary definition of the dipole moment} \quad (52)$$

$$d \left(t - \frac{r}{c} \right) = q \vec{x}_{ret} \quad (53)$$

$$\implies P(t) = \frac{2}{3c^3} \left(|\ddot{d} \left(t - \frac{r}{c} \right)|^2 \right) \quad (54)$$

$$(55)$$

To find the velocity of the charge required to have relativistic corrections larger than 5% we can examine two cases when $\vec{\beta} \perp \dot{\vec{\beta}}$, and $\vec{\beta} \parallel \dot{\vec{\beta}}$.

For the former

$$P(t) = \frac{2}{3} \frac{q^2}{c} \left[\gamma^6 \left(\dot{\vec{\beta}}^2 - \left(\vec{\beta} \times \dot{\vec{\beta}} \right)^2 \right) \right]_{ret} \implies P(t) = \frac{2q^2}{3c} \left[\gamma^6 \left(\dot{\vec{\beta}}^2 - \vec{\beta} \dot{\vec{\beta}}^2 \sin^2 \theta \right) \right]_{ret} \quad (56)$$

$$\rightarrow \frac{2q^2}{3c} \left[\gamma^6 \dot{\vec{\beta}}^2 \left(1 - \vec{\beta}^2 \right) \right]_{ret} = \frac{2q^2}{3c} \left[\gamma^4 \dot{\vec{\beta}}^2 \right]_{ret} \quad (57)$$

Here we see the relativistic correction is due to the γ^4 . Thus we set $\gamma^4 = 1.05$. ($n = 4$) Here I can derive the correction for a general quantity that is proportional to γ^n

$$\frac{1}{(1 - \beta^2)^{\frac{n}{2}}} = 1.05 \quad (58)$$

$$1 - \beta^2 = \frac{1}{(1.05)^{\frac{2}{n}}} \text{ thus } \implies \beta = \sqrt{1 - \frac{1}{(1.05)^{\frac{2}{n}}}} \rightarrow \sqrt{1 - \frac{1}{(1.05)^{\frac{2}{4}}}} = 0.155241512 \quad (59)$$

For the latter case we have

$$P(t) = \frac{2q^2}{3c} \left[\gamma^6 \left(\dot{\vec{\beta}}^2 - \vec{\beta} \dot{\vec{\beta}}^2 \sin^2 \theta \right) \right]_{ret} \rightarrow P(t) = \frac{2q^2}{3c} \left[\gamma^6 \dot{\vec{\beta}}^2 \right] \quad (60)$$

So the correction is given by ($n = 6$)

$$\beta = \sqrt{1 - \frac{1}{(1.05)^{\frac{2}{6}}}} \rightarrow \sqrt{1 - \frac{1}{(1.05)^{\frac{2}{6}}}} = 0.1270112325 \quad (61)$$

Thus we can see that once the speed reaches 15.5% of the speed of light in the $\perp$ case we get a correction of 5%. For the $\parallel$ case if the speed is 12.7% the speed of light we get a correction of 5%. We see that the $\parallel$ case is the lower bound where we are in the relativistic regime, while the $\perp$ is the upper bound i.e. if we analyze the $\perp$ case and find it to be relativistic then all the other possible directions must also be relativistic. So we need the velocity to be less than $0.127c$ to be in the non relativistic regime as defined by 5% correction to γ ($\forall$ possible directions).

Question 3

We have the formula for the relativistic doppler formula given by

$$\frac{\lambda_o}{\lambda_s} = \frac{\nu_s}{\nu_o} = \sqrt{\frac{1+\beta}{1-\beta}} \quad (62)$$

where λ_o is the observed wavelength and λ_s is the source wavelength, and ν is the corresponding frequencies. We know that the wavelength of visible green light is $550nm$ and the wavelength of visible red light is $700nm$. Thus for the red light to appear green then

$$\sqrt{\frac{1+\beta}{1-\beta}} = \frac{700}{500} \approx 1.28 \quad (63)$$

$$(1.28)^2 (1-\beta) = 1+\beta \implies \beta (-1.28^2 - 1) = 1 - 1.28^2 \quad (64)$$

$$\implies \beta = 0.2419648272 \quad (65)$$

$$v = 0.2419648272c = 72589448.16ms^{-1} = 261322013.37598km/h \quad (66)$$

Since the speed is above the typical speed limit of $120km/h$ he was arrested.